

GROUP ASSIGNMENT

Instructions

- This is a **group assignment**.
- Each group must discuss and answer **all questions**.
- **Submission Deadline – Presentation - 24th June 2026**

Question 1

A newly launched mobile banking application experiences frequent system crashes, transaction failures, and customer complaints within its first month of operation. The bank's management is concerned about financial losses and damage to its reputation.

Required:

- a) Define Software Quality Assurance (SQA).
- b) Explain three objectives of SQA that could help address the issues in this scenario.
- c) Discuss three consequences of poor software quality for the bank.
- d) Explain how SQA contributes to customer satisfaction.

Question 2

A software development company discovers that defects reported by customers cost significantly more to fix than defects identified during requirements analysis. Management wants to invest more in quality assurance activities.

Required:

- a) Explain the concept of the Cost of Poor Quality.
- b) Describe three reasons why early defect detection is important.
- c) Identify three SQA activities that can help prevent defects during development.
- d) Explain the difference between SQA and Software Testing.

Question 3

A software company is evaluating a new healthcare management system and wants to assess its maintainability, portability, reliability, and usability before deployment.

Required:

- a) Explain the purpose of the ISO/IEC 25010 quality model.
- b) Describe the four quality characteristics mentioned in the scenario.
- c) Explain how quality models support software quality measurement.
- d) Discuss two challenges of measuring software quality.

Question 4

The national revenue authority plans to replace its manual tax filing process with an Online Tax Declaration and Payment System. The system will be used by thousands of taxpayers and government officials. Since tax regulations are defined by law, the requirements are stable and unlikely to change.

Required:

- a) Recommend the most suitable SDLC model for this project.
- b) Explain three characteristics of the selected model that make it suitable for this project.
- c) Describe three QA activities that should be conducted during the Requirements phase.
- d) Explain why early defect detection is important in this project.

Question 5

A software startup develops a food delivery application using Scrum. Despite releasing software every two weeks, customers continue to report defects in completed features.

Required:

- a) Explain the role of the Definition of Done (DoD) in Scrum.
- b) Describe how Test-Driven Development (TDD) improves quality.
- c) Explain the role of QA during Sprint Planning and Sprint Review.
- d) Recommend two actions the team can take to reduce defects.

Question 6

A school introduces a management system to handle student records, fee payments, and examination results. During User Acceptance Testing (UAT), teachers discover that grade calculations do not match the school's grading policy.

Required:

- a) Explain whether this is a Verification or Validation issue.
- b) Differentiate between Verification and Validation.
- c) Describe two Verification activities and two Validation activities that should have been conducted.

Question 7

A hospital introduces an online patient management system. Before deployment, the QA team must ensure that all modules work correctly together and that the complete system meets user requirements.

Required:

- a) Differentiate between Integration Testing and System Testing.
- b) Explain why both testing levels are important in this scenario.
- c) Identify two defects that Integration Testing could detect.
- d) Explain the role of Acceptance Testing before deployment.

Question 8

An online shopping platform announces a **48-hour Black Friday Mega Sale**, offering discounts of up to 70% on electronics, clothing, and household products. The company expects approximately 50,000 concurrent users, but no large-scale performance tests are conducted before the event. Within the first hour of the sale, customers experience slow page loading times, failed login attempts, shopping carts that do not update correctly, and payment transactions that timeout. As traffic continues to increase, the website becomes completely unavailable for several hours.

Required:

- a) Identify the most appropriate non-functional testing category that should have been conducted before the sale and justify your answer.
- b) Differentiate between Load Testing and Stress Testing.
- c) Explain how both testing techniques could have helped prevent the failure.
- d) Discuss two business consequences of the system failure.

END